When Does the Second View Help? Auditing Cross-View Fusion in Urban Vision-Language Models

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Abstract. Pairing overhead satellite and ground-level street imagery is widely assumed to give vision–language models a richer view of a place than either alone. We test when this assumption holds. We build OELLM, a deterministic, OpenStreetMap-grounded benchmark that pairs a marked satellite tile with road-aligned street views across 40 cities, spanning urban-attribute tasks (answerable from one location) and cross-view tasks (answerable only from the relationship between views). Our instrument is a four-way image ablation—blind, satellite-only, street-only, and both—from which we read *fusion synergy* (the gain of both views over the better single one) and an item-level *interference asymmetry*. On seven urban-attribute tasks, a fine-tuned 9B VLM gains only +0.5 points from the second view, and at the item level it overturns a correct answer about nine times as often as it rescues a wrong one, so a per-item single-view oracle beats the two-view model by 12 points. On five cross-view tasks the same model and instrument gain +19 to +71 points, confirming the audit detects fusion when the task requires it. The pattern replicates across model families and scales, stronger prompting, inference-time view pruning, and training-time view count. Multi-view fusion is task-shaped, not pervasive.

Keywords: Geosensing · Vision-language models · Multi-view fusion

1 Introduction

More perspectives are assumed to be better: a satellite tile shows layout and connectivity, a street-level view shows facades, surfaces, and signage, and pairing them should give a vision–language model a richer picture of a place than either alone. A growing line of work collects, aligns, and jointly models satellite and ground views for urban understanding on exactly this premise. But "should" is rarely tested directly—existing benchmarks report one accuracy over a mix of tasks, some answerable from a single view and some requiring both, so a high number can mean genuine fusion or simply that one view sufficed. We ask the question the aggregate hides: *when does the second view actually help?*

We answer it with a controlled audit. We build OELLM, a deterministic, OpenStreetMap-grounded benchmark that pairs a marked satellite tile with road-aligned street-level views across 40 cities, spanning urban-attribute tasks (answerable from one location) and cross-view tasks (answerable only from the relationship between views). Our instrument is a four-way image ablation—blind, sat, sv, full—from which we read *fusion synergy* (full − max(sat, sv), the gain of both views over the better single one) and an item-level *interference asymmetry*. We evaluate at the minimum-sufficient input—one satellite tile plus one forward street view—so every comparison sits on the same regime, and apply the audit to a fine-tuned 9B VLM and a panel of raw open VLMs. Figure 1 previews the setup and the headline effect.

The result is a sharp dissociation. On the seven urban-attribute tasks, the second view adds only +0.5 points over the better single view on average, and at the item level it is mildly *harmful*: it overturns a correct answer about nine times as often as it rescues a wrong one, so a per-item single-view oracle—an upper bound on single-view routing—beats the two-view model by 12 points ($p < 10^{-60}$). On the five cross-view tasks the same model and instrument gain +19 to +71 points, so the audit registers fusion exactly when the task requires it. The pattern is robust: it reproduces across raw VLMs of different families and scales, survives a 16k-token reasoning budget and a multi-turn self-check, holds when inference-time pruning keeps a single street view (which retains 89% of correct answers), and persists when the model is trained with all four street views instead of one (which adds only ∼2 points).

We frame this carefully. It is a *task-dependent* observation about single-view-decidable attribute recognition at commonly served satellite resolutions, not a claim that overhead imagery is uninformative or that fusion fails in general—the cross-view controls show large synergy on the same models. We report bootstrap confidence intervals on every estimate, audit the benchmark for text-only leakage, and characterise the items where the second view changes the answer.

Our contributions are:

1. **A task-dependence result for cross-view fusion.** On seven urban-attribute tasks a fine-tuned 9B VLM gains only +0.5 points from a second view over the better single view, whereas on five cross-view tasks built into the same benchmark it gains +19 to +71—a within-model dissociation showing that pairing helps only when the task structurally requires it.
2. **An item-level interference analysis.** Beyond the small average, the second view is net-harmful on the urban tasks: it overturns a correct answer about nine times as often as it rescues a wrong one, and a per-item single-view oracle exceeds the two-view model by 12 points—evidence that the model adjudicates poorly between redundant views rather than failing to read them.
3. **OELLM, a deterministic paired-view benchmark** of 12 OSM-grounded task families over 40 cities, with a four-way image ablation and built-in cross-view controls that make the single-view-sufficient versus fusion-required distinction measurable.
4. **A reusable audit protocol**—fusion synergy, vision contribution, and interference asymmetry—for deciding, before committing to dual-view collection, whether a second view is warranted; we show the verdict is robust across

model families and scales, inference-time view pruning, and training-time view count.

2 Related Work

Remote-sensing vision-language models are single-view. A wave of instruction-tuned VLMs target remote sensing, including GeoChat [6], SkySenseGPT [15], LHRS-Bot and LHRS-Bot-Nova [9,17], EarthGPT [26], and the multi-sensor generalist EarthDial [19]. These models are overhead-only: each consumes a single aerial or satellite image, and none ingests ground-level imagery (Tab. 2). Where "multi-modal" appears here it refers to multiple *overhead* sensors—optical, SAR, infrared—rather than to pairing satellite with street-level views. The dominant suites follow suit: RSVQA [14], VRSBench [8], and GeoChat's own benchmark pose questions over a single overhead image. None isolates the marginal value of a second, ground-level view. Architecturally these build on LLaVA-style designs [11, 12] with CLIP or SigLIP encoders [18, 25]; we use four as zero-shot baselines and discuss their training-data provenance in Sec. 4.1.

Multi-view urban benchmarks. A smaller set of benchmarks pairs overhead and ground views and notices the surface phenomenon we study, but stops short of explaining it. UrBench [28] reports that multi-image and single-image performance differ little on coordinate-aligned street/satellite pairs, but attributes the gap to question *comprehension* and measures only aggregate accuracy, leaving open whether the second view is unused, redundant, or actively harmful. CityLens [13], concurrent with our work, finds street view alone competitive with both views—but on a three-task socioeconomic *regression* target, with a single model and no item-level or cross-view analysis. Neither isolates *where in input-mode space* the effect lives. We contribute the missing instrument: a four-way image ablation yielding a per-item interference asymmetry and a single-view oracle, applied across a task taxonomy whose built-in cross-view controls confirm the same instrument registers large fusion when fusion is required. The result is not "a second view rarely helps"—already suspected—but a *mechanism* (the model overturns about nine correct answers for each it rescues) and a *boundary* (fusion is decisive exactly when the answer lives between views). Convergent prior observations make this audit timely; they do not make it redundant. UrbanLLaVA [3] trains over satellite, street-view, OSM, and trajectory inputs but poses its attribute tasks from satellite alone; CityCube [23] targets cross-view spatial reasoning rather than attribute recognition.

Cross-view geo-localization: where both views are necessary. Satellite–ground pairing is well established in cross-view geo-localization, where the goal is to match a ground photo to an overhead reference [2, 24, 29]. There the two views are jointly *necessary*, and methods succeed by modeling the distinct-but-linkable layout they share [27]. This is the regime our cross-view controls occupy;

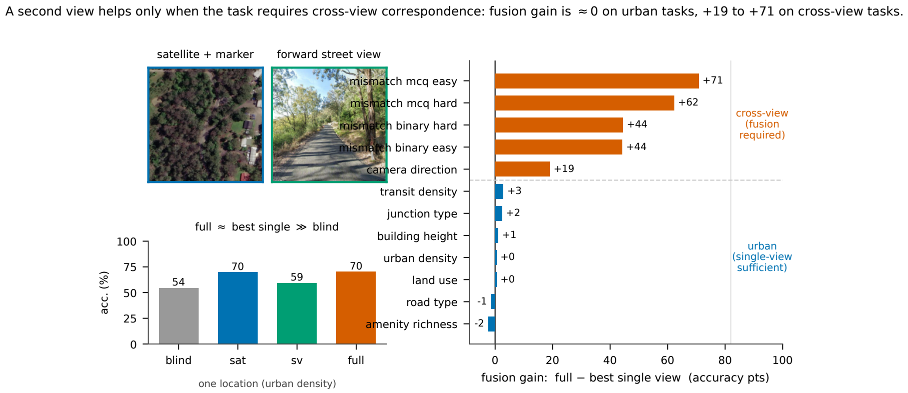

Fig. 1: Left: one held-out location (Sydney). Each benchmark location provides a marked sub-meter satellite tile and four cardinal street-level views; the dataset always supplies all four. Our primary model consumes only *two* images—the satellite tile and the *forward* street view—and the ablation is computed in that configuration: "full" here means satellite + forward street view. For this task (urban density) full barely exceeds the better single view, while both far exceed the no-image prior (blind). **Right:** per-task fusion gain (full− better single view) across all 12 evaluated tasks. The second view helps only when the task requires cross-view correspondence: ≈0 on the seven urban tasks (some negative), +19 to +71 on the five cross-view tasks. Imagery is unmodified dataset content: satellite tile © Esri World Imagery; street-view panels © Google (Street View).

complementarity that is constitutive of localization is largely absent from single-view-decidable attribute recognition.

The marginal value of a second modality. That adding a modality need not help is known beyond geosensing. Wang *et al.* [21] show the best single-modal network can outperform a naively fused one, and Liang *et al.* [10] formalize multimodal interaction into redundancy, uniqueness, and synergy, showing low-synergy regimes are real; our "fusion synergy" is the behavioral analogue of their synergy term. Against this, multimodal learning can provably lower achievable risk [5], and several aerial+ground systems report recognition gains [16,22]. Most directly counter to our setting, Suel *et al.* [20] find fusing satellite and street imagery improves *regression* of income and deprivation; we engage this in Sec. 6, noting it differs in task type (regression vs. four-way recognition), labels, and model. Our finding refines rather than contradicts the "fusion helps" literature by naming a regime where it does not.

3 The OELLM Benchmark

We build **OELLM**, a deterministic, OpenStreetMap-grounded benchmark that pairs overhead and street-level imagery to make the question *"when does a second*

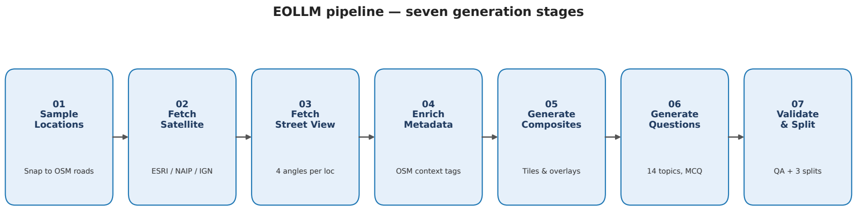

Fig. 2: The seven-stage deterministic generation pipeline. Every stage reads and enriches the same list of sample dictionaries; no human annotation is involved.

view help?" measurable. Every answer derives from OSM tags or coordinate geometry rather than human annotation, so the corpus is bit-exact reproducible and carries no annotator drift. We describe the construction, the task taxonomy, the splits, and the evaluation protocol, and state the trade-off honestly: the benchmark is limited to what OSM and geometry can certify, but every answer is verifiable and the same OSM snapshot regenerates the same questions.

3.1 Construction

Each *location* is sampled within one of 40 cities across 32 countries on six continents and is associated with (i) one satellite tile, (ii) four street-level views at fixed road-aligned bearings (forward, backward, left, right), and (iii) OSM-derived attribute labels. A seven-stage pipeline (Fig. 2) threads each sample through location sampling, satellite-tile fetching, street-view capture, OSM enrichment, composite rendering, deterministic question generation, and validation/splitting.

Satellite imagery is sourced per location by geography—sub-meter aerial orthophotos: ESRI World Imagery ($\sim$0.3 m/px to 0.5 m/px in cities) as the global default, NAIP ($\sim$0.6 m/px) in the US, and IGN ($\sim$0.2 m/px) in France, with a coarse Sentinel-2 fallback ($\sim$10 m/px) reserved for locations no high-resolution source covers. Across the 34,484-record corpus, 31,489 questions (91%) draw ESRI imagery, 1946 NAIP, and 1049 IGN; the Sentinel-2 fallback was never triggered, so the redundancy we report later is measured against uniformly sub-meter overhead imagery rather than a coarse proxy. The satellite tile carries a marker indicating the queried location; the marker is part of the input and is held fixed across all ablation modes (its effect is examined in Sec. 5). Street-level views are retrieved from Google Street View and retain Google's original attribution; satellite imagery is © its respective provider (Esri, NAIP/USDA, IGN).

3.2 Task taxonomy

The benchmark contains 12 task families in two groups (Fig. 3). Two satellite-only tasks present in earlier drafts, `green_space` and `road_surface`, were *retired* from the released benchmark because every model we evaluated solved them almost perfectly and they therefore carried no discriminative signal; we discuss this audit in Sec. 5 and report on the 10 retained tasks below.

Urban understanding (seven tasks). land_use, building_height, urban_density, road_type, junction_type, amenity_richness, and transit_density. Each is a four-way multiple-choice question whose answer follows from one location's OSM context; the visual is a marked satellite tile plus street-level imagery. These are the tasks for which "do we need both views?" is well-posed, because each is in principle answerable from imagery of a single location.

Cross-view geolocalisation (five tasks). camera_direction asks which of four arrows on the satellite tile points in the heading a single street view was captured from; mismatch_binary_{easy,hard} ask whether a street-view set belongs to a satellite tile; mismatch_mcq_{easy,hard} ask which of four street-view composites matches the tile. *Easy* variants draw distractors from another city (where vegetation, signage language, and vehicle types often leak the answer); *hard* variants draw them from the same city, defeating those shortcuts and isolating the geometric match. These answers live in the *relationship* between views and are unanswerable from a single view by construction.

A fair test of fusion. The two groups differ in what fusion can buy, and this difference is the benchmark's design principle. An urban-attribute question is, by construction, answerable in principle from *either* modality, and empirically each single view alone carries the answer: for most urban tasks the satellite-only and street-only accuracies of our trained model track each other closely (e.g. land_use 72.4 vs. 71.7, building_height 60.0 vs. 62.6, road_type 72.7 vs. 76.0; Sec. 4.2). The question is therefore not whether one view *can* suffice but whether a model *gains* from holding both—a fair test of fusion precisely because each view is independently sufficient. The cross-view group is the complement: a class of tasks that genuinely *require* fusion, included so the same instrument can register large synergy when fusion is the only route to the answer. We do not claim urban semantics can never require cross-view reasoning; we claim that the attribute tasks deployed in current urban-VLM benchmarks do not, and we provide the cross-view group to exhibit tasks that do.

3.3 Splits

All splits operate at the location level, so no location's questions span two splits. We hold out a disjoint *benchmark* of 4734 questions (all 12 released tasks) drawn from a mixture of seen and unseen cities, shipped in two parallel files (`public`, no answers; `with_answers`, gold-labelled). For training we use two strategies that share this benchmark (Tab. 1): a *per-city* split (random locations per city to validation) and a *seen/unseen* split (eight cities held out entirely, probing geographic generalisation). Our primary model is trained on the seen/unseen split. Question generation seeds option shuffling from `sample_id` + topic, so the

Table 1: Record counts for the released OELLM corpus (34,484 questions over 4168 locations in 40 cities). Two training strategies share one disjoint benchmark. This paper's fusion audit uses the seven urban-attribute tasks of the benchmark ($n = 2629$); the remaining 2105 cross-view questions form the positive control of Sec. 4.3.

Split set	Train	Validation	Benchmark
per_city	23,431	6319	—
seen_unseen	23,282	8179	—
benchmark (12 tasks)	—	—	4734

correct letter is stable across re-runs but distributed across the corpus; we keep one question per (location, task) pair to avoid location-level inflation, and discard locations with fewer than four street-level views.

3.4 Evaluation protocol

We evaluate every model under a controlled *image ablation*: each question is posed under four input conditions—blind (question and options only), sat (the marked satellite tile only), sv (street-level view only), and full (both). The text of the question and options is byte-identical across conditions; only the supplied images change. We decode greedily with a fixed seed and parse a single answer letter with a strict word-boundary matcher; the same harness, prompt, and rendering run across all models so that only the forward pass differs. Of the 4734 benchmark questions spanning all 12 released tasks, this study's fusion audit uses the 2629 questions from the seven urban-attribute tasks; the remaining 2105 cross-view questions serve as the positive control. Per-task question counts are uneven by design, reflecting OSM-tag availability rather than importance (building_height, 190; junction_type, 334; the other five urban tasks, 421 each).

3.5 Coverage and limitations

We flag two forms of uneven coverage. Geographically the corpus is Europe-weighted (the single largest share of records), with the remaining cities spread across five other continents; absolute accuracies should be read with that skew in mind, though our headline metric is computed within each question and is unaffected by the geographic mix. Label-wise, a few attribute tasks depend on OSM tags that are sparse in some cities—`building:levels`, which underlies building_height, is sparse in São Paulo and Singapore—so those tasks contribute fewer questions. Full distributional statistics are provided in the accompanying dataset report.

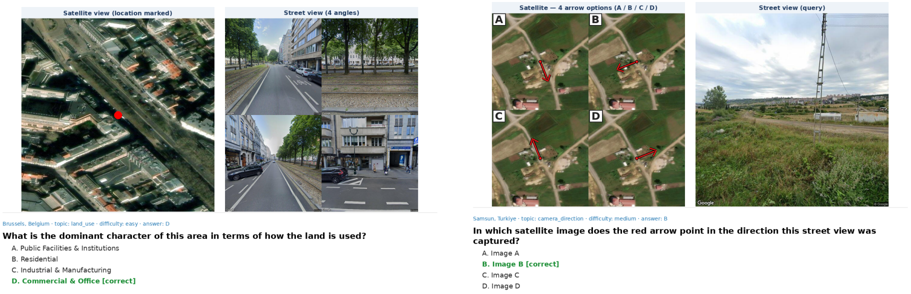

Fig. 3: Two rendered tasks. **Left** (urban understanding, land_use): a marked satellite tile and four street views; the answer follows from one location's OSM context. **Right** (cross-view, camera_direction): four satellite crops with candidate heading arrows and one street-view query; the answer lives in the relationship between the views.

Table 2: Off-the-shelf remote-sensing VLMs evaluated zero-shot on the overhead (satellite-only) half of the benchmark. All four are single-image and *overhead-only*; none ingests ground-level imagery. "OSM data" marks whether the model's training corpus is derived from OpenStreetMap (relevant to the provenance caveat in Sec. 5).

Model	Base LLM	Vision encoder	Input res.	OSM data
GeoChat-7B [6]	Vicuna-1.5-7B	CLIP ViT-L/14	504	No
SkySenseGPT-7B [15]	Vicuna-1.5-7B	CLIP ViT-L/14	336	No
LHRS-Bot-Nova [9]	Llama-3-8B	SigLIP-L/14	336	**Yes**
EarthDial-4B-RGB [19]	Phi-3-Mini	InternViT-300M	tiled	**Yes**

4 Experiments

4.1 Setup

Our primary model is **Qwen3.5-9B** fine-tuned with LoRA [4] (rank 16, $\alpha = 16$, all linear layers including the vision tower, bfloat16, max image edge 512 px) on the seen/unseen training split. We train it at the *minimum-sufficient* input—a marked satellite tile plus the single *forward* street view (satfwd), two images per sample—so that every comparison sits on the same one-street-view regime. Fine-tuning on the task distribution is the cleanest setting for the view question: a failure to use both views cannot then be blamed on poor task alignment. For the mismatch tasks, which structurally require comparing the full street-view set against the satellite tile, evaluation feeds all four street views even to this adapter, because the task design forces it.

We evaluate under the four image-ablation conditions of Sec. 3.4: blind, sat (marked tile), sv (forward street view), and full (both). In this configuration full means *two* images—one satellite, one street view; we report the four-street-view training regime separately in Sec. 4.8. We report per-task accuracy and two derived quantities: *fusion synergy* $=$ full $-$ $\max(\text{sat}, \text{sv})$, the gain of both views

Table 3: Trained Qwen3.5-9B (satfwd) on the held-out benchmark, per urban task and input condition (accuracy, %). **Syn.** = full − max(sat, sv) is fusion synergy; **Vis.** = full − blind. Synergy is small on every task; full stays close to the better single view, while vision contribution is large.

Task	n	blind	sat	sv	full	Syn.
land_use	421	66.5	72.4	71.7	72.9	+0.5
building_height	190	46.8	60.0	62.6	63.7	+1.1
urban_density	421	54.4	69.6	59.1	70.1	+0.5
road_type	421	68.6	72.7	76.0	74.6	−1.4
junction_type	334	48.2	71.6	66.5	74.0	+2.4
amenity_richness	421	24.5	55.3	47.7	53.0	−2.4
transit_density	421	28.7	46.3	46.1	49.2	+2.9
Mean	—	48.3	64.0	61.4	65.3	+**0.5**

over the better single one, and *vision contribution* = full − blind. To check that the finding is not model-specific we also evaluate four off-the-shelf, overhead-only remote-sensing VLMs zero-shot (Tab. 2; Sec. 4.5) and a panel of raw open VLMs under the same four conditions (Sec. 4.6); all run through one harness on a single 96 GB GPU.

4.2 The second view adds little over the better single view

Table 3 reports the trained model per urban task under all four conditions. Two facts stand out. First, full tracks the *better single view* closely: per-task fusion synergy ranges from −2.4 (amenity_richness) to +2.9 (transit_density) and averages +0.5 points across the seven tasks; two tasks have *negative* synergy, and street view alone already matches or exceeds full on road_type. Pooled over all 2629 items, synergy is +1.1 points with a paired bootstrap 95% interval of $[-0.2, +2.4]$ that includes zero; a two-one-sided-test (TOST, [7]) against a ±3-point margin declares the effect equivalent to zero. The second view does not measurably improve accuracy on these tasks. Second, vision still matters: full − blind is large (for example +20 on transit_density and +29 on amenity_richness), so the model reads the imagery—it simply extracts about a single view's worth of usable information from it.

Whether this small margin reflects genuine fusion or the model leaning on its stronger view is resolved at the item level (Sec. 4.4). But first we confirm the measurement is sensitive: that a synergy near zero means the second view is unused, not that our instrument cannot see fusion.

4.3 The same model fuses when the task requires it (positive control)

Could the small attribute-task synergy reflect an insensitive measurement rather than a property of the tasks? Table 4 runs the same model and harness on the

Table 4: Positive control: the same trained model on cross-view tasks that require both views by construction (the street-view set is fed in full). Synergy is large, confirming the ablation detects fusion when it is needed.

Cross-view task	blind	sat	sv	full	Syn.
camera_direction	25.2	23.0	–	42.0	+19.0
mismatch_binary_easy	45.1	47.0	53.4	97.6	+44.2
mismatch_binary_hard	56.8	48.0	44.2	92.4	+44.4
mismatch_mcq_easy	23.5	26.1	23.8	96.9	+70.8
mismatch_mcq_hard	23.5	22.6	24.9	87.2	+62.2

five cross-view tasks, whose answers require comparing the two views. Synergy is large, +19 to +71 points: mismatch_mcq_easy rises from 26% with the satellite alone to 97% with both views, and camera_direction gains +19. The instrument that reads a half-point gain on the attribute tasks reads a forty-to-seventy-point gain when the task is built to require both views.

We are careful about what this control does and does not show. It establishes that the model and the instrument *can* register large synergy when fusion is the only route to the answer—so a synergy near zero on the urban tasks is genuine absence, not measurement failure. It is *not* an image-count-matched comparison to the urban setting: the cross-view tasks feed the full street-view set (five images) by construction, whereas the urban full condition uses two. The image-count-controlled question—does the second *modality* help when added to the first?—is answered within the urban tasks themselves (Secs. 4.2 and 4.4), where full adds exactly one street view to one satellite tile. Two further facts keep the control honest: the null on the urban tasks is not a dead satellite channel, since the identical adapter gains +44 to +71 points here; and an independent, purpose-built cross-view retrieval network (Sample4Geo [1]), transferred zero-shot, scores 0.64 on the binary same-place questions—above the 0.50 chance rate—confirming the matching tasks carry genuine cross-view signal that the attribute tasks do not offer to recover (a different model class and metric, not comparable to Tab. 4).

4.4 Item-level: the second view interferes more than it helps

A small positive average could still hide genuine fusion on a subset of items. We join the four conditions' predictions at the item level—every question carries a stable identifier shared across conditions—and analyse the 2629 fully paired urban items directly (Tab. 5 and Fig. 4).

The second view overturns far more correct answers than it rescues. Genuine fusion is rare: full is correct while *both* single views are wrong on only 40 items (1.5%). Interference is common: full is wrong while at least one single view was right on 363 items (13.8%). The second view therefore overturns a correct answer about nine times as often as it rescues a wrong one. Equivalently—and this is

Table 5: Item-level analysis on the 2629 paired urban questions. **Fusion-win** = full correct while *both* single views fail; **Interfere** = full fails while a single view succeeded. The oracle keeps the better single-view answer per item (an upper bound on single-view routing, not an achievable system); **full−oracle** carries a paired bootstrap 95% CI. The second view rescues few items and breaks many more.

full	oracle	full−oracle	Fusion-win	Interfere	ratio
65.2	77.5	-12.3 $[-13.7, -10.9]$	1.5%	13.8%	9.1:1

the *same* statistic stated as an upper bound—an *oracle* that keeps the better of the two single-view answers per item (a router, not a fusion model, and not an achievable system, since selecting the right view uses the gold label) beats full by 12.3 points (77.5 vs. 65.2; paired bootstrap 95% CI $[10.9, 13.7]$; McNemar $p < 10^{-60}$). The two-view model is beaten by a single-view router on every one of the seven tasks (Fig. 4). Against the street-only view, full is significantly *worse* than parity ($p < 10^{-4}$); against the satellite-only view the win/loss split is not significant (174 vs. 145, $p = 0.12$). When the two views disagree, the model does not reliably keep the correct one; the extra view is, on net, a distractor.

This is not a leakage artefact, and stronger reasoning does not remove it. The oracle gap is the same on the items that genuinely require vision: 12.3 points over all items and 12.2 points restricted to the 1357 items the model answers incorrectly while blind. Nor is the obstacle a lack of deliberation. A natural objection is that the second view is complementary but the model simply fails to combine it; if so, stronger inference-time reasoning should surface the latent signal. It does not: across plain decoding, a 16k-token explicit-reasoning budget, and a four-turn observe–reason–self-check–commit protocol, mean accuracy rises modestly while fusion synergy remains *negative* and street-view-alone stays the single strongest condition (evaluated on an AWQ-quantised Qwen3.5-9B so the long-context runs fit one GPU; this probe is an internal trend check, not numerically comparable to the headline, and the claim rests only on the sign of synergy, which quantisation does not flip). More deliberation improves how the model reads one view, not its ability to fuse two—consistent with the picture that the obstacle is adjudicating between redundant views.

The average gain, precisely. We are careful not to overclaim in either direction. The pooled synergy is $+1.1$ points with a 95% interval $[-0.2, +2.4]$ that includes zero, equivalent to zero under a ± 3-point TOST; the effect is, at most, a couple of points and is not statistically distinguishable from no effect. The *harm*, by contrast, is significant: on the items the second view changes, it removes 14% of correct answers while rescuing 1.5%, and the single-view oracle's 12.3-point lead has an interval well clear of zero. Whether a near-zero average gain at that interference cost warrants serving a second imagery modality is the practical question we take up in Sec. 6.

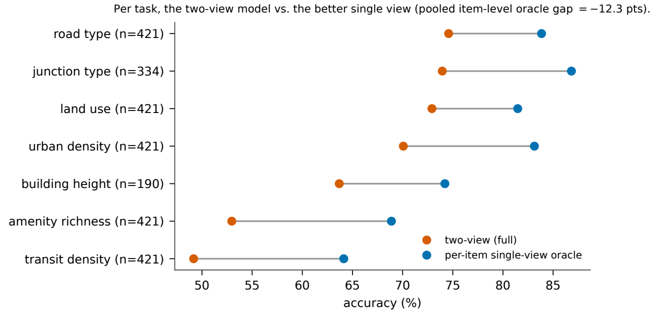

Fig. 4: Per urban task, the two-view full condition versus a per-item single-view oracle that keeps the better single-view answer for each question (an upper bound on single-view routing, not an achievable system). The oracle leads on every task; the pooled gap is -12.3 points. Trained Qwen3.5-9B (satfwd).

Table 6: Off-the-shelf RS-VLMs, zero-shot, single marked satellite tile, unweighted mean accuracy over the seven urban tasks (each model's per-task scores averaged over the identical task set; green_space and road_surface excluded as in Sec. 4.2). Last column: our trained model in satellite-only mode, the like-for-like single-overhead-view reference. †trained on OSM-derived text; read as an upper bound (Sec. 5).

	GeoChat	SkySenseGPT	LHRS-Nova†	EarthDial†	*Ours (sat)*
Urban-7 mean	21.0	27.2	29.5	37.8	***64.0***

4.5 The finding holds zero-shot across off-the-shelf RS-VLMs

To rule out that the redundancy is an artefact of our particular model, we evaluate four off-the-shelf remote-sensing VLMs zero-shot on the satellite-only half of the urban benchmark (Tab. 6). None was trained on our task, our four-way format, or our marker overlay; they span CLIP- and SigLIP-based encoders, three LLM families, and single-crop versus tiled preprocessing. This comparison is not a claim of superiority—a model fine-tuned on the distribution should beat zero-shot baselines—and it does not bear on the synergy result, which is a within-model difference. It establishes the single-overhead-view floor: the best zero-shot model (EarthDial, 37.8%) trails our trained model's satellite-only accuracy (64.0%) by 26 points, and LHRS-Bot-Nova and EarthDial, trained on OSM-derived text, should be read as an upper bound on visual skill (Sec. 5).

4.6 The regime split reproduces across families and scales

The same four-mode audit on raw, zero-shot open VLMs—Qwen3.5-4B/9B, Qwen2.5-VL-7B, and Gemma-4-12B—preserves the *shape* of the result. The raw models are weaker overall (urban full accuracy clusters near 50% across all four, versus 65% for our trained model), but the pattern is the same: a small urban fusion gain and a large cross-view fusion gain. On the raw Qwen3.5-9B, for instance, urban synergy is essentially zero (and slightly negative on land_use and junction_type), while mismatch_mcq_easy jumps from 25% with the satellite alone to 66% with both views. We evaluate the raw panel under each model's native street-view input scheme rather than our single-forward-view regime; because these models are not fine-tuned and view economy is not the question we ask of them, the panel speaks only to whether the qualitative regime split recurs, not to a like-for-like view-count comparison with our adapter. If anything this grants the raw panel more street-view evidence to fuse, so the near-zero urban synergy it still exhibits is a conservative reading. The regime split is thus a property of current VLMs and current paired-view tasks, not a fine-tuning artefact.

4.7 Inference-time view pruning

If a single street view carries the urban signal, pruning the others at inference should cost little. We measure this on a stronger raw model (Qwen3.6-27B) by the fraction of *already-correct* answers that survive when the street-view set is reduced to k images under different selection rules. Among questions the model answers correctly with all views, keeping only the forward street view retains 89%, and keeping none (satellite only) retains 67%—a survival rate, not an accuracy. Informed view selection beats random selection only modestly (top-attention 90% vs. random 85% at k=1), and the 67% zero-view floor reflects residual text and label leakage, so the evidence that any *specific* single view is essential is weak. The reading consistent with Sec. 4.2: one street view supplies most of what four do for these tasks.

4.8 What do additional street views buy at training time?

Finally we ask whether training with all four street views, rather than one, changes the picture. We compare our satfwd adapter against an otherwise identical model trained on the marked satellite tile plus all four street views (5-STV), same backbone and seen/unseen split, each evaluated in its own full mode. On the seven urban tasks the 5-STV model averages 67.6% versus satfwd's 65.3%—a 2.3-point gain from training with three additional street views, well below the marginal cost of carrying four times the visual data. On the cross-view tasks the two are comparable. For urban-attribute deployment, the forward view is most of the value; the additional street views are not what the model learns from.

5 Robustness and Confounds

We check the confounds a templated benchmark and a within-model ablation invite.

Option text and position. Because the questions are templated, we verify the answer cannot be read off the options. On the seven urban-attribute tasks that carry our headline result, the correct letter is uniform (A/B/C/D = 662/694/628/645; $\chi^2 = 3.6$, $p = 0.31$ against uniform; an always-'A' baseline scores 25.2%), and the correct option is not systematically the longest—its mean length (25.9 characters) is slightly below that of the distractors (26.5), and it is the single longest option on only 22.9% of items, below the 25% chance rate. We note for completeness that across the full 12-task corpus the letter distribution is skewed (always-'A' 29.3%), an artefact of the option-shuffling RNG documented in the dataset report; this skew is carried by the cross-view tasks (always-'A' 34.4%) and is absent from the urban tasks evaluated here. Crucially, our headline metric is a within-item comparison across image conditions (full vs. a single view) on identical option text, so any residual option regularity cancels in the difference and cannot produce the effect.

Text-only leakage and degenerate tasks. We treat full − blind as the vision-attributable signal and audit every task in blind mode. This audit removed two tasks from the released benchmark: green_space, which a model answered at 100% accuracy with no image (the correct option wording was identifiable independent of its position), and road_surface, which was near-ceiling and essentially text-solvable. Removing degenerate tasks before reporting is a strength, not a gap: it leaves the seven urban tasks, on each of which the imagery contributes a meaningful margin over the text-only prior (for example +22 points of vision contribution on transit_density and +29 on amenity_richness; Sec. 4.2). The interference result of Sec. 4.4 is, moreover, robust to any residual leakage: the per-item single-view oracle beats the two-view model by 12.3 points overall and by 12.2 points when restricted to the 1357 items the model answers *incorrectly* while blind—the items that genuinely require vision. No model refused or hedged on any urban item in any mode.

Image count. A concern in multi-view ablations is that one condition may simply present more images than another. In our primary configuration this concern does not arise: the sat, sv, and full conditions supply one, one, and two images respectively, so the comparison of full against either single view adds exactly one image of the other modality, with no token-count asymmetry between the two views. (The cross-view control tasks of Sec. 4.3 feed the full street-view set by construction; we treat them as a sensitivity check on the instrument, not as an image-count-matched comparison, and the headline result does not depend on them.)

OSM provenance. Our labels are OSM-derived, and two of the four zero-shot remote-sensing models we compare against—LHRS-Bot-Nova and EarthDial—are trained on OSM-derived text, so their scores may reflect shared OSM priors; we treat them as an upper bound on visual skill and mark them accordingly

(Sec. 4.5). This does not touch our main result, which is a *within-model* difference (full vs. the better single view): any view-independent OSM prior cancels in the difference.

Satellite resolution. The redundancy is measured against high-resolution overhead imagery, not a coarse proxy: every benchmark tile resolves to a sub-meter source (ESRI $\sim$0.3 m/px to 0.5 m/px, NAIP $\sim$0.6 m/px, IGN $\sim$0.2 m/px), and the Sentinel-2 fallback is never triggered. For tasks that are structurally ground-level no overhead resolution recovers the cue. We cannot rule out that still higher ground sampling distance would help the few tasks where overhead geometry is plausibly resolution-limited (building height, urban density), and flag this as the main open question (Sec. 6.1).

6 Discussion

Our reading is that complementarity is a property of the task, not of imagery in general. An attribute question—local land use, building height, transit access—can be answered from one location's imagery, and the two views then encode much the same answer-relevant content, so the model reads it off whichever view is convenient and is, on net, distracted by the other. A cross-view question (whether a street photo belongs to a tile, or which way a camera points) instead depends on the relationship between the views, where our models do use both. The seven attribute tasks fall in the first regime, the cross-view tasks in the second.

We frame the mechanism behaviorally and decline to overclaim it. Our measurements show the model derives little usable signal from the second view on these tasks and is, on net, distracted by it; we do not claim to identify the internal cause. Redundant information across views, a representation dominated by one modality, and an architecture that fails to combine them would all produce the pattern we observe, and our data do not distinguish among them. What we establish is the input–output fact—adding the second view does not help and often hurts—which is what a practitioner deciding whether to collect it needs to know.

This has a practical consequence. Street-level and high-resolution satellite imagery differ sharply in cost, coverage, and licensing, yet for these attribute families the second view adds at most a couple of points and degrades about one item in ten. A deployment can serve a single view—the cheaper or better-covered one—at little accuracy cost; and when both are available, the evidence favors *routing* each question to one view over feeding the model both. The synergy, vision-contribution, and interference measures we use form a reusable instrument for making that call before committing to dual-view collection.

What we do and do not claim. We claim that current VLMs do not, on these urban-attribute tasks, convert a second view into accuracy, and that the second view is on net a mild distractor; we do not claim VLMs are incapable of fusion—the cross-view controls show the same models fusing when fusion is the only route to the answer. We claim the second view's information is largely redundant *for these tasks at the resolutions we serve*; we do not claim satellite imagery is uninformative, nor that no task or higher ground sampling distance would benefit. Our title is a question we answer conditionally: the second view helps when the task structurally requires cross-view correspondence, and largely does not when a single location's attributes are at issue.

6.1 Limitations and scope

The claim concerns seven urban-attribute tasks in four-way MCQ form at commonly served satellite resolutions; it is not a claim that overhead imagery is uninformative or that fusion fails in general, as the cross-view controls show large synergy on the same models. Three caveats bound it further: our two training regimes share one held-out test set, of which the held-out-cities split is the true generalization check; higher ground sampling distance might help the few overhead-geometry-limited tasks (building height, urban density); and an open-ended or regression formulation could surface complementary signal an MCQ cannot express.

7 Conclusion

We measured, rather than assumed, the value of a second perspective for urban attribute recognition. Across model scale, training regime, inference-time view count, and prompting strategy, the second view adds at most a couple of points over the better single view and overturns a correct answer about nine times as often as it rescues a wrong one, while the same models gain 19–71 points on cross-view tasks that require both views. Multi-perspective necessity is thus task-dependent. We release OELLM—benchmark, adapters, and the audit protocol—so that the next paired-modality study can ask, before collecting a second view, whether its tasks are the kind that need one.

8 Future Work

Because only the cross-view tasks consistently benefit from multiple street views, the natural next step is to exploit those views for richer geometric reasoning. The four road-aligned angles per location contain the signal needed for local 3D reconstruction; lifting urban understanding from 2D image fusion to 3D spatial reasoning would let a model reason about occlusion, scale, building geometry, and pedestrian-environment affordances in ways single 2D images cannot support, with OELLM's four-angle captures as a natural input substrate. Further directions include fusion architectures that genuinely combine modalities rather than selecting between them, scaling to more cities to reduce the Europe-heavy distribution, and extending deterministic OSM grounding to socio-economic indicators.

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